

INAD Prediction Metrics and CGFM Comparison Summary

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This note gives a compact, presentation-oriented summary of prediction metrics, the Henderson-Shimakura Table 4 reproduction, and the bolus rolling one-step prediction comparison. The main comparison is between integer-valued antedependence (INAD), negative-binomial generalized linear model (NB GLM), negative-binomial generalized linear mixed model (NB GLMM), and correlated gamma frailty model (CGFM) predictors.

1 Supplement To Last Week's Validation Summary

1.1 Marginal-Moment MARD Table

This section adds the CGFM time-varying frailty row to the bolus marginal moment MARD table from last week's validation summary. MARD means mean absolute relative discrepancy, reported as a percentage; lower is better.

The NB GLM and CGFM independent frailty models have the same one-time negative-binomial predictive family under the same time and group linear predictor. The table therefore uses a single NB GLM / CGFM independent row based on the reproduced Henderson-Shimakura independent-frailty moments.

Model	G1 mean	G1 var	G1 var excl.	G2 mean	G2 var
C-alpha INAD	8.35	26.74	15.65	9.31	22.79
U-INAD	5.92	29.05	20.37	5.83	19.72
Poisson GLM	9.95	59.70	64.84	5.96	78.52
NB GLM / CGFM ind.	8.94	42.44	29.53	6.87	20.75
NB GLMM	8.35	58.35	43.73	8.24	24.25
CGFM shared	9.86	29.01	23.14	6.01	41.83
CGFM time-varying	8.96	42.26	29.70	6.88	21.08

Adding the CGFM time-varying row does not change the table’s main reading: the INAD class still gives the best MARD in the table, with unconstrained INAD best for G1 mean, G2 mean, and G2 variance, and constrained-alpha INAD best for G1 variance.

2 Henderson-Shimakura Table 4 Reproduction

Henderson and Shimakura (2003), Table 4, was reproduced from the bolus data to check the CGFM reference models.

2.1 Model-Level Checks

Model	Reproduced LL	Paper LL	Key parameter check
Independent frailty	-2191.77	-2191.77	$\beta = 0.34$; $\theta = 0.49$; information SE and robust SE match printed values.
Shared frailty	-2249.16	-2249.16	$\beta = 0.33$; $\theta = 0.24$; SEs match printed values.
Time-varying frailty	-2159.65	-2159.5	$\beta = 0.3428$, $\theta = 0.4907$, $\rho = 0.8487$; parameter estimates match Table 4 to printed precision.

The full time-varying fit converged from data and recovered $\rho \approx 0.85$, so the paper’s reported correlation parameter is reproducible rather than merely paper-calibrated.

The likelihood column verifies reproduction within each model row; it is not a cross-model likelihood comparison. The independent and shared rows are full log-likelihoods, whereas the time-varying row is the normalized pairwise composite log-likelihood used by Henderson and Shimakura. The time-varying objective value is close to, but not an exact one-decimal reproduction of, the paper value (-2159.65 from data versus -2159.5 reported).

2.2 Time-Varying Row Reproduced From Data

Parameter	Script estimate	Script SE	Paper Table 4
alpha1	2.1043	0.10	2.10 (0.10)
alpha2	1.6164	0.13	1.62 (0.13)
alpha3	1.7118	0.12	1.71 (0.12)
alpha4	1.7898	0.11	1.79 (0.11)
alpha5	1.9664	0.10	1.97 (0.10)
alpha6	1.6621	0.10	1.66 (0.10)
alpha7	1.4741	0.13	1.47 (0.13)
alpha8	1.3701	0.12	1.37 (0.12)
alpha9	1.3743	0.11	1.37 (0.11)
alpha10	1.5398	0.11	1.54 (0.11)
alpha11	1.3910	0.10	1.39 (0.10)
alpha12	1.3479	0.12	1.35 (0.12)
beta	0.3428	0.13	0.34 (0.13)
theta	0.4907	0.06	0.49 (0.06)
rho	0.8487	0.03	0.85 (0.03)

3 Prediction Metrics

Let Y_i be the observed count for prediction case i , let $\hat{m}_i = E(Y_i \mid \mathcal{H}_i)$ be the predictive mean, and let $\hat{p}_i(y) = P(Y_i = y \mid \mathcal{H}_i)$ be the predictive probability mass function conditional on the available history $\mathcal{H}_i$. Let $\hat{F}_i(k) = \sum_{y \leq k} \hat{p}_i(y)$ be the predictive CDF. RPS is the primary metric for the bolus comparison below. Log score is the second distributional score, and RMSE is kept as a point-forecast check.

3.1 RPS

In words. RPS, the ranked probability score, measures the full predictive CDF against the realized count. For count data, it sums the squared distance between the predictive CDF and the step function induced by the observed count. Lower RPS is better. Compared with log score, RPS is less driven by a single tail probability and is often more stable for discrete empirical PMFs.

Formula.

For prediction case i , let the finite scoring support be $k = 0, 1, \dots, K_i$. Then

$$\text{RPS}_i = \sum_{k=0}^{K_i} \left\{ \hat{F}_i(k) - \mathbf{1}(Y_i \leq k) \right\}^2.$$

The reported RPS is the average over prediction cases:

$$\text{RPS} = \frac{1}{n} \sum_{i=1}^n \text{RPS}_i.$$

In this validation work, K_i is not a single global maximum. It is a case-specific finite support chosen to include the observed value and the predictive support used for that model's scoring calculation. For simulation-based INAD forecasts this means the support extends to the maximum of the observed value and the simulated predictive samples. This finite-support choice is why the support/smoothing caveat matters for RPS and log score.

3.2 Log score

In words. Log score measures how much probability the predictive distribution assigns to the value that actually occurs. It is a proper probabilistic scoring rule: a model is rewarded for assigning high probability to the observed count and penalized for putting too little mass there. Lower log score is better: the score is defined as $-\log \hat{p}$, so smaller values mean the model placed more probability on the observed count. For simulation-based empirical PMFs, log score can be sensitive to tail smoothing because it evaluates the PMF at one point.

Formula.

$$\text{LogScore} = -\frac{1}{n} \sum_{i=1}^n \log \hat{p}_i(Y_i).$$

For an individual prediction case,

$$\text{LogScore}_i = -\log \hat{p}_i(Y_i).$$

3.3 RMSE

In words. RMSE measures point-forecast accuracy. It compares the predictive mean with the observed count, squares the error, averages over prediction cases, and then returns to the original count scale by taking the square root. Lower RMSE is better. Because errors are squared, RMSE penalizes large misses more heavily than MAE.

Formula.

$$\text{RMSE} = \left\{ \frac{1}{n} \sum_{i=1}^n (\hat{m}_i - Y_i)^2 \right\}^{1/2}.$$

4 Additional Prediction Metrics

Prediction comparisons usually use a mix of point, distributional, and calibration metrics. The following metrics are useful context but are not the main bolus summary metrics reported below.

Metric	Type	What it measures	Useful here?
MAE	Point forecast	Mean absolute error, $\frac{1}{n} \sum_i \hat{m}_i - Y_i $.	Redundant with RMSE for the present summary; useful as a robustness check.
PIT / randomized PIT	Calibration	Whether predictive distributions are calibrated; well-calibrated PIT values should be approximately uniform on $[0, 1]$.	Useful as a diagnostic plot or calibration check, but not used as a one-number headline metric here.
Brier score	Probabilistic	Squared error for binary or categorical event probabilities.	More natural for categorical outcomes than count outcomes.
Interval coverage	Calibration	Whether nominal prediction intervals cover at the advertised rate.	High-value future supplement once coverage is computed consistently for all displayed models.
Interval score	Probabilistic / interval	Rewards narrow intervals with correct coverage and penalizes misses.	High-value future supplement if prediction intervals are added.
CRPS	Probabilistic	Continuous analogue of RPS.	Useful robustness check if a continuous approximation is desired.
Dawid-Sebastiani score	Probabilistic moment score	Uses predictive mean and variance.	Possible supplement, but less informative than full-PMF scores for counts.

In plain language, PIT asks whether the predictive distribution is calibrated: after randomization for discreteness, observations should fall throughout their predictive distributions as if they were uniform draws on $[0, 1]$. A PIT histogram or interval-coverage table would be the preferred calibration display; the present summary does not use PIT as a one-number comparison.

For the bolus comparison, the reported metric set is **RPS (primary), log score, and RMSE**. RPS and log score evaluate the predictive distribution; RMSE records whether the distributional gains come with comparable point accuracy. Ad hoc folded diagnostics are omitted because they are not proper scores and can be hard to interpret as prediction evidence.

5 Bolus Rolling One-Step Prediction Comparison

Design: the bolus data comprise 65 patients in two treatment groups of size 30 and 35. Each fold holds out one patient from each group and fits on the remaining 63, so the all-pairs leave-one-per-group-out design gives $30 \times 35 = 1050$ folds. Prediction is rolling one-step at $t = 2, \dots, 12$.

The NB GLM row is aligned with the independent-CGFM reference: under the same time and group linear predictor, the independent Poisson-gamma frailty model integrates to the same negative-binomial predictive family as the NB GLM.

The t -statistics are computed after aggregating to held-out patients. This is more conservative than treating all 1050 all-pairs folds as independent, because each patient appears in many folds.

For this comparison, RPS is treated as the primary prediction metric: it is a proper distributional score for count-valued predictive CDFs and is less sensitive than log score to one observed tail probability. Log score is the second distributional score, and RMSE is the point-forecast check.

The table reports paired differences:

$$\Delta = \text{score}(\text{constrained-alpha INAD}) - \text{score}(\text{reference}).$$

Negative values mean constrained-alpha INAD has the lower, better score. The number in parentheses is the patient-level paired t -statistic. Here, constrained-alpha INAD denotes the INAD fit that keeps the thinning parameter α constant across time.

Model	Delta RPS (t)	Delta RMSE (t)	Delta log score (t)
Unconstrained INAD	-0.012 (-0.6)	-0.028 (-0.9)	-0.006 (-0.7)
Poisson GLM	-0.081 (-2.9)	-0.013 (-0.4)	-0.181 (-5.4)
NB GLM	-0.405 (-3.9)	-0.627 (-4.2)	-0.124 (-3.3)
NB GLMM	-0.158 (-2.2)	-0.164 (-1.7)	-0.059 (-1.99)
CGFM shared	-0.072 (-1.1)	0.015 (0.2)	-0.144 (-3.2)
CGFM time-varying	-0.066 (-2.3)	0.020 (0.5)	-0.017 (-1.3)

The absolute score scale is shown below for context. Lower values are better; RMSE is averaged at the held-out-patient level, matching the paired comparison above.

Model	RPS	RMSE	Log score
Constrained-alpha INAD	2.207	3.870	2.673
Unconstrained INAD	2.219	3.898	2.679
Poisson GLM	2.288	3.883	2.854
NB GLM	2.612	4.497	2.797
NB GLMM	2.365	4.034	2.732
CGFM shared	2.279	3.855	2.817
CGFM time-varying	2.273	3.851	2.690

The NB GLM row also represents the independent-CGFM predictive family under the same time and group linear predictor. RPS and log-score values are sensitive to finite-support and smoothing conventions; this sensitivity is most relevant for the closest RPS comparisons, especially CGFM time-varying. The much larger RPS gain over NB GLM is least exposed to this issue; the Poisson and NB GLMM RPS gains should still be read with the support-convention caveat in mind.

5.1 Reading The Table By Metric

RPS. This is INAD’s strongest broad metric. Constrained-alpha INAD is indistinguishable from unconstrained INAD and better than Poisson GLM, NB GLM, NB GLMM, and CGFM time-varying by the conventional $|t| > 2$ threshold. It is directionally but not significantly better than CGFM shared. If the primary RPS family of reference comparisons is adjusted conservatively for multiplicity, the strongest RPS wins, especially NB GLM and Poisson, remain the most secure; the NB GLMM and time-varying CGFM RPS gains should be read as supportive rather than standalone proof.

Log score. Constrained-alpha INAD is comparable to unconstrained INAD. It clearly beats Poisson GLM, NB GLM, and CGFM shared. Against NB GLMM, the log-score advantage is borderline ($t = -1.99$, approximately $p = 0.05$). It is essentially tied with CGFM time-varying.

RMSE. Constrained-alpha INAD is comparable to unconstrained INAD, Poisson GLM, CGFM shared, and CGFM time-varying. It clearly beats NB GLM and is directionally better than NB GLMM but not quite at the conventional threshold. The defensible point-forecast statement is therefore comparable point accuracy, not uniformly better point prediction.

6 Summary Conclusion

The current evidence supports the following prediction statement: INAD’s advantage is in probabilistic distributional one-step prediction, not in uniformly better point forecasts.

1. **INAD’s clearest real-data advantage is probabilistic distributional prediction, especially RPS.** It improves RPS over Poisson GLM, NB GLM, NB GLMM, and time-varying CGFM, with the strongest support against NB GLM; the CGFM shared comparison is directionally favorable but not significant.
2. **The distributional evidence is coherent against the simpler count-regression baselines.** Against Poisson and NB GLM, INAD improves both RPS and log score. Against NB GLMM, the RPS result is clearer than the borderline log-score result.

3. **Constrained-alpha and unconstrained INAD are practically tied.** The constrained-alpha fit is slightly better on RPS, RMSE, and log score, but none of those differences is statistically meaningful; it remains the primary row because it is the validation parameterization used throughout.
4. **CGFM time-varying is the toughest comparator.** INAD has the better RPS, but ties on RMSE and log score. CGFM shared is also close on RPS. Until a common-support sensitivity check is fully documented, the closest RPS margins should be framed as supportive rather than overclaimed.
5. **Point-error metrics do not support a universal prediction win.** RMSE shows comparable point accuracy outside the NB GLM comparison. This result should be acknowledged rather than hidden.

This is a one-dataset, rolling-one-step validation on the bolus data; recursive multi-step prediction and fully standardized support/smoothing sensitivity for RPS and log score remain follow-up checks rather than settled evidence here.